

Graduation Project Proposal

Personal Decision-Style Engine

A Personalized Web Based Human Decision Mirror system using LLM's

An AI system that learns how a specific person decides — and advises based on who they are, not what's objectively optima

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Abstract

Most advice and recommendation systems are organized around a single question: what is the objectively best option? That framing works reasonably well for products, routes, or media, where "best" can be judged against a shared external standard. It works far less well for the decisions that actually shape a person's life, such as which job to accept, whether to end a relationship, or how to handle a conflict with family, because there is no external standard for these choices, only the individual's own values, risk appetite, and priorities. A recommendation can be well-reasoned and still feel wrong for the person receiving it, simply because it was not built around who that person is.

This project, referred to as The Decision Mirror, proposes a different approach. Rather than identifying the best available option, the system estimates which choice is most consistent with how a specific person has made decisions in the past. A profile of the user's decision-making style is built from two sources: free-text narratives of past decisions, written in the user's own words, and a short onboarding quiz used to establish an initial profile before enough narratives have accumulated. Both inputs are processed by a large language model that extracts a structured set of trait scores, covering dimensions such as risk tolerance, time horizon, and the balance between emotional and analytical reasoning. This profile is updated continuously, with recent decisions weighted more heavily than older ones.

When a user brings a new decision to the system, the response is not generated in a single step. The system retrieves the user's trait profile together with their most relevant past decisions, asks follow-up questions where the situation is ambiguous, and only then produces a recommendation, along with an explanation of which part of the user's own history it is drawing on. This conversational structure is intended to separate the project from simpler retrieve-and-respond designs.

To keep the central claim testable despite the subjective nature of the problem, the system is evaluated by withholding each user's most recent logged decisions and checking whether a model built from their earlier history predicts those withheld choices more accurately than a generic, non-personalized baseline. The intended contribution of this work is not a general-purpose advice engine, but evidence that a person's decision-making style can be extracted from unstructured personal narratives and used to produce guidance that is measurably closer to that individual than advice generated without such context.

1. Problem Statement

Existing decision-support and recommendation systems are almost always built to converge on a single "correct" or "popular" answer, computed the same way regardless of who is asking. This is a reasonable design when options are comparable across users, but it breaks down for personal, high-stakes decisions, where the right answer for one person is the wrong one for another with different priorities. An early version of this project attempted to sidestep the problem by working only with structured decision data, such as chess move prediction, where every option is already machine-readable. That version was technically convenient but did not reflect the actual problem: real decisions about careers, relationships, and money do not arrive with a pre-defined list of options the way a chess position does, so a system built only for structured data would not generalize to real use.

The problem this project addresses is therefore twofold: first, how to model an individual's decision-making tendencies from the kind of messy, unstructured information people actually produce about their own choices, and second, how to use that model to support a live decision through something closer to a conversation than a lookup, without pretending that a single *correct* answer exists for personal decisions in the first place.

2. Proposed Solution

The Decision Mirror removes the requirement for structured decision data entirely. Instead of needing machine-readable options and choices, it accepts free-text decision narratives and uses a large language model to perform the structuring work that would otherwise require a rigid schema. The system is built around three ideas that distinguish it from a generic chatbot wrapped around a language model:

- Personality-fit over correctness — the system is not trying to identify the objectively best choice; it is trying to identify the choice most consistent with the user's own demonstrated decision style.
- A quantifiable trait profile — free-text logs and quiz responses are converted into a structured trait vector per user, rather than being fed back into a prompt as raw text on every request.
- Conversational advisory rather than one-shot output — when a live decision is presented, the system asks clarifying questions before producing a recommendation, instead of answering immediately from a single retrieval step.

3. Objectives

- Design and implement a two-channel data collection mechanism: open-ended logging of past decisions, and a structured onboarding quiz for new users.
- Build an LLM-based extraction layer that converts free-text logs and quiz responses into a structured, quantifiable decision-style trait vector.
- Implement recency-weighted aggregation so that a user's running profile reflects recent behavior more strongly than older or quiz-derived signal.
- Implement a conversational advisory loop that retrieves a user's profile and relevant past decisions, asks follow-up questions, and produces a personalized, explained recommendation.
- Design and run a held-out evaluation protocol comparing personalized predictions against generic and majority-vote baselines.

4. System Overview

Data about the user is built from two complementary channels that feed the same profile:

- Channel 1 — Free-form decision logging: the user records past situations and the choices they made, in their own words, at any time, with no fixed schedule or rigid form.
- Channel 2 — Onboarding psychological quiz: a short, structured quiz completed once at signup. The LLM analyzes the answers to derive an initial read on the user's decision-making style, giving the system a usable profile from day one instead of waiting for organic logs to accumulate.

Both channels are processed through the same extraction layer and merged into a single running profile:

- Extract — every free-text log and quiz response is run through an LLM-based extraction step that pulls structured trait signals out of it.
- Aggregate — trait signals from both channels accumulate into one running profile, weighted so recent decisions count more than old ones, and quiz-derived priors are progressively diluted as real logged decisions accumulate.

When the user faces a live decision, the loop becomes a conversation rather than a single lookup:

- Present — the user brings the live decision and the options they are weighing.
- Retrieve — the system retrieves the user's trait profile and the most relevant past decisions by semantic similarity, not just recency.
- Converse — the LLM asks clarifying or follow-up questions where the situation or profile leaves something ambiguous, rather than immediately producing a verdict.
- Advise — once enough context exists, the system reasons over the user's pattern and the new situation and helps the user reach a decision, surfacing its reasoning rather than only a verdict.

5. The Decision-Style Trait Layer

Pasting a user's past decisions directly into a language model's context window on every request would be prompt engineering rather than a research contribution. The intended contribution is an extraction layer that converts each free-text log and each quiz response into a structured trait vector, covering candidate dimensions such as:

- Risk tolerance — whether the choice favors the safe, known option or the uncertain, higher-upside one.
- Time horizon — short-term relief versus long-term payoff orientation.
- Locus of priority — self-interest versus the interests of others or a relationship.
- Decisiveness speed — quick and intuitive versus deliberated and researched.
- Emotional versus analytical weighting — how much the stated reasoning leans on feeling versus logic and data.
- Consistency — how often a user's choices align with their own previously stated values, tracked as a self-consistency score over time.

Each new log or quiz answer nudges the user's running profile vector. This profile is the artifact that is evaluated, visualized, and reused at inference time.

6. Cold-Start Handling

A new user with only two or three logged decisions cannot get a meaningful profile from logging alone. This is addressed directly by Channel 2: a set of forced-choice scenario questions, similar in spirit to a personality inventory, is analyzed by the LLM immediately after signup, bootstrapping an initial trait profile within minutes. Organic free-text logging then refines this profile and, over time, increasingly outweighs the quiz-derived prior. This is treated as a required component rather than an optional addition, since without it the system would be unusable for a new user's first weeks.

7. Evaluation Methodology

The evaluation plan is intended to make an otherwise subjective claim testable, and was designed early rather than left for later:

- Held-out prediction test — a user's most recent N logged decisions are hidden, the system predicts each from the earlier history and quiz-derived profile, and the predictions are checked against what the user actually chose.
- Personalized versus generic baseline — the personalized model's accuracy is compared against a generic, no-history model and a majority-vote baseline. The core hypothesis is that the personalized model outperforms both.

- Profile stability check — the user's trait vector is tracked over time to see whether updates shrink as more data accumulates, which would suggest the extraction layer is converging on a real signal rather than noise.
- Optional longitudinal validation — where possible, users are followed up with after a real decision to check whether they acted as predicted; this is the strongest but hardest evidence to collect.

8. Data Collection Strategy

How to collect enough real decisions with enough context, when real life does not log itself the way a game does, was the central design problem this project had to solve. The plan rests on the two channels described in Section 4:

- Primary channel — organic logging: free-text logging is the ongoing data collection mechanism, not a separate research instrument; every user generates training and evaluation data simply by using the product.
- Bootstrap channel — onboarding quiz: forced-choice scenarios double as an initial labeled dataset across users, useful for calibrating the extraction layer and for solving cold-start per user.
- Volume strategy — a small initial cohort (classmates, friends, online communities) will use both channels for several weeks before the conversational advisory feature is evaluated.
- Noise control — because logging is voluntary and the quiz is a fixed, deliberately designed instrument, only decisions the user considered worth recording, or a structured proxy for their style, enter the dataset, avoiding the low-signal problem of passively mined behavioral data.

9. System Architecture

Layer	Role
Logging interface	Lightweight chat-style or form-style input where users freely describe a past decision whenever they want (Channel 1).
Onboarding quiz module	A short forced-choice psychological instrument completed once at signup (Channel 2); answers are analyzed by the LLM to bootstrap an initial trait profile.
Extraction layer	LLM-based component that converts each free-text log and each quiz response into structured trait scores (Section 5); the core NLP/ML contribution of the project.
Profile store	Running, recency-weighted trait vector per user, merging quiz-derived priors and organic log signal, plus the raw log history for retrieval.
Retrieval layer	Given a live decision, retrieves the user's most semantically relevant past decisions by embedding similarity rather than only the most recent ones.
Conversational advisory layer	LLM prompted with the user's trait profile, retrieved similar past decisions, and the live decision; holds a back-and-forth conversation, asking follow-up questions as needed, before producing a personalized, explained recommendation.
Evaluation harness	Held-out prediction testing and baseline comparisons (Section 7), run offline against accumulated logs and quiz data.

10. Risks and Limitations

- Pattern reinforcement risk — if a user has a track record of, for example, conflict avoidance, and a new situation genuinely calls for confrontation, a purely descriptive system risks reinforcing an unhealthy pattern rather than helping. The system is designed to be descriptive ("here is what your pattern suggests") rather than prescriptive ("here is what you should do"), and this distinction is communicated to the user directly, not only in the written report.
- Self-report bias — logged decisions and quiz answers reflect how users narrate their own choices, not an objective record, so the profile models how a person describes their decisions as much as how they actually decide. This is treated as a disclosed limitation rather than a fatal flaw.
- Cold-start dependency — even with the onboarding quiz, the system is weaker in a user's first few weeks, before enough organic logs exist to refine the quiz-derived prior.
- Evaluation ceiling — human decisions are not perfectly predictable even by people who know the individual well, so a meaningful outcome is "better than baseline," not high absolute accuracy, and expectations are framed accordingly.

11. Work Plan

- Build the free-text logging interface and the onboarding quiz module first, since no modeling is possible without data.
- Build the trait-extraction layer and run it against quiz responses and early organic logs; this is the first technical milestone and a natural midterm checkpoint.
- Build the retrieval layer and the conversational advisory layer once enough per-user history exists for recommendations to be meaningful.
- Build the evaluation harness in parallel with the previous step, since held-out testing needs to run continuously as data accumulates rather than being added at the end.
- Recruit a small external cohort beyond the team's own usage as early as possible, since cross-person generalization is one of the most important claims to check.

12. Conclusion

The Decision Mirror is proposed as a system that learns a person's decision-making fingerprint from the choices they have already made and how they describe those choices, and that uses this fingerprint in a live conversation to help the user reach a decision that fits who they are, while making explicit which part of their own history the recommendation is drawn from. The project's central claim, that personalized modeling outperforms generic advice on a person's own held-out decisions, is designed to be testable rather than asserted, and the risks and limitations of the approach are treated as part of the contribution rather than left unstated.